P1: Assignment 1.2 : Linux 101 Hands-on

CY5001 — Linux 101 Hands-On Assignment

Objective

In this lab, you will connect individually to your assigned Ubuntu Linux container via SSH and complete a series of basic Linux tasks.
There is no report to submit.

Connection Instructions

You will receive an email with your connection credentials.

This is what you'll be receiving in your email:

-A **username** (e.g. sierra.jos, your actual username will be provided by email)

- A **password** (e.g. u8e4106347bb1b060 your actual password will be provided by email)

- A **custom SSH port number** (e.g. 22222, your actual password will be provided by email)

Example SSH Command Format:
ssh sierra.jo@cy5001p1.mscy.org -p 22222

You must:

- Connect **only** to your assigned container.

- Log in **only** with your assigned username and password.

Lab Rules

- Complete all work **inside your home directory** (/home/your_username).
- Any text file you create **must include your own username** in its contents.
- **Do not install unauthorized software** or store unrelated data.
- Your system has **limited resources**; overloading it may destroy your work.
- Follow the exact naming and content instructions.

IMPORTANT All tasks results Must be stored under /home/sierra.jo/linux101/

Task List

Task 1. Session Recording with asciinema

- Install the asciinema package.
- Start a recording session: asciinema rec linux101.cast
- Perform all subsequent tasks while the recording is running.
- When done, press Ctrl+D to finish recording.
- **After finishing, play back the cast** to verify it recorded correctly:asciinema play linux101.cast
- Also, ensure that your "**linux101.cast**" file is copied to "**/home/cyuser??/linux101/**" when the recording is done

Commands to use: apt-get install asciinema, asciinema rec, asciinema play

Task 2. Basic Navigation and File Operations

- Create a directory called linux101.
- Inside linux101, create a text file called welcome.txt that contains your full name in the line: CY5001 Assignment 1.2: Your first and Last name
- Copy welcome.txt to a new file called assignment1.2.txt.
- Create a subdirectory called practice.
- Move assignment1.2.txt into the practice/ directory.

Commands to use: pwd, mkdir, nano, cp, mv, cat, ls, echo

Task 3. File Permissions

- Change the permissions of welcome.txt so that it is **read-only** for all users.

Commands to use: chmod, ls -l

Task 4. Installing Software

- Update the package database.
- Install the package htop.

Commands to use: apt-get update, apt-get install htop

Task 5. Processes and Jobs

- Start a background sleep process: sleep 1209600 &

- Verify that the sleep 1209600 process is running.

Commands to use: ps, htop ,grep

Task 6. User and Group Management

- Create a new user called testuser. (any password you can remember)
- Create a new group called testgroup.
- Add testuser to the group testgroup.

Commands to use: adduser, groupadd, usermod, getent

Task 7. Downloading Files and Archiving

- Use wget to download the homepage from:https://users.cs.duke.edu/~ola/ap/linuxwords into your linux101directory.
- The name of your downloaded file needs to be linuxwords
- .Create a compressed archive of your linux101 directory named linux101.tar.gzin your home folder.

Commands to use: wget, tar

Task 8. System Information Collection

- Inside linux101, create a file named sysinfo.txt that contains:
    - The current date (date command)
    - The current user (whoami)
    - The hostname of your container (hostname)
    - Your First and Last name

Commands to use: date, whoami, hostname, echo

Task 9. Shell Redirection and Logs

- Inside linux101, redirect the output of ls -la into listing.txt.
- Also redirect any error output to an specific error file errors.txt. To trigger those errors you need to run the following command "find / -type f -name 'linuxwords' without using administrator privileges.
- The errors.txt must contained only the errors but not the result of the command.

Commands to use: ls, >, >>, 2>, 2>>, echo

Task 10. Installing and Using tree

- Install the tree package.
- Inside linux101, run:tree
- Save the output into structure.txt and include your username in the file
    - If your e-mail is smith.m@northeastern.edu, then your username is "smith.m"

Commands to use: apt-get install tree, tree, >, echo

Summary of Deliverables

By the end of the lab, you must have the following files and configurations:

No.	Item Type	Description	Task Reference
1	File	linux101.cast session recording present and operational in linux101/	Task 1
2	File	welcome.txt with proper content in linux101/	Task 2
3	File	practice/assignment1.2.txt exists and contains the correct content	Task 2
4	File	welcome.txt has read-only for owner, group and others	Task 3
5	Installed Software	htop and tree are installed	Tasks 4 & 10
6	Background Process	sleep 1209600 is running	Task 5
7	User/Group	testuser exists and is a member of testgroup	Task 6
8	File	linuxwords properly downloaded into linux101/	Task 7

No.	Item Type	Description	Task Reference
9	Archive	linux101.tar.gz archive created in your home directory and extracts correctly to linux101/	Task 7
10	File	sysinfo.txt includes date, user, and hostname	Task 8
11	File	listing.txt and errors.txt generated with corresponding content	Task 9
12	File	structure.txt generated by tree command	Task 10